MILITARY SPECIFICATION

NUT, PLAIN, WELDING,
GENERAL SPECIFICATION FOR, METRIC

METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metrically dimensioned steel plain welding nuts (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging and Packing for Shipment and Storage.

MILITARY

MIL-I-6868 - Inspection Process, Magnetic Particle.
DOD-N-63476/1 - Nut, Plain, Welding, Metric.

STANDARDS

FEDERAL

FED-STD-H28/21 - Screw-Thread Standards for Federal Services, Section 21, Metric Screw Threads.
FED-STD-66 - Steel: Chemical Composition and Hardenability.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5310

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1312 - Fasteners, Test Methods.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) HANDBOOK

SAE J122A-1975 - Surface Discontinuities of Nuts.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. Recycled and reclaimed materials shall be used to the maximum extent practicable. Nuts shall be made of weldable carbon steel conforming to FED-STD-66, composition 1005 through 1020 or 1108 through 1110, capable of withstanding a minimum proof load stress of 800 MPa.

3.3 Hardness. Nut hardness shall be not greater than 30 HRC.

3.4 Protective finish. Nuts shall be furnished uncoated.

3.5 Design.

3.5.1 Threads. Nut threads shall conform to the "M" profile, tolerance class 6H, as specified on the applicable specification sheet and as defined in FED-STD-H28/21.

3.5.2 Bearing surface. The nut bearing surface shall be perpendicular to the thread axis within 3 degrees.

3.5.3 Pilot. Thread and pilot axes shall be concentric within 3 percent of the pilot diameter.

3.6 Dimensions. Dimensions and tolerances shall be in accordance with the applicable specification sheet and as specified herein.



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3.7 Discontinuities. The depth of discontinuities such as laps, seams and inclusions in the nuts shall be not greater than 3 percent of the nominal thread size. The length of discontinuities shall not exceed the height of the nut.

3.7.1 Cracks. A crack is a clear crystalline fracture passing through or across the grain boundaries without inclusion of foreign elements. Quench cracks are not permitted in any location. Forging cracks located in the top and bottom face are permitted provided: no more than two forging cracks extend from the thread across the full width of the face; no forging crack extends into the thread beyond the first full thread; and the width of any forging crack does not exceed 0.20 mm plus 0.01D, where D is the nominal nut thread size in mm. Forging cracks located at the intersection of the top or bottom face with a flat are permitted provided that the width does not exceed 0.25 mm plus 0.020D, where D is the nominal nut thread size in mm. The depth of forging cracks shall be not greater than 3 percent of the nominal nut thread size in mm.

3.8 Proof load. Nuts shall withstand the proof loads specified in table I without rupture of metal or thread stripping. Upon removal of the proof load, nuts shall be manually removable from a mating bolt or mandrel without assistance of a tool except for the initial one-half turn of the nut.

TABLE I. Proof loads.

| Thread size | M3 | M4 | M5 | M6 | M8 | M10 | M12 | M14 | M16 |
|---|---|---|---|---|---|---|---|---|---|
| Proof load, kN (min) 1/ | 4 | 7 | 11.4 | 16 | 29 | 46 | 67 | 92 | 126 |
| Stress area ($mm^2$) | 5.03 | 8.78 | 14.2 | 20.1 | 36.6 | 58 | 84.3 | 115 | 157 |

1/ Proof load is based on a nut material proof stress of 800 MPa, min. and the stress area of the mating bolt or mandrel thread.

3.9 Workmanship. Finished nuts shall be uniform in quality and appearance. Surfaces shall be free of foreign matter, burrs, loose scale and any other defects which may adversely affect serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Materials inspection (see 4.4).
b. Quality conformance inspection (see 4.5).



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4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed at ambient temperature, humidity and atmospheric pressure.

4.4 Materials inspection. Materials inspection shall consist of certification supported by verifying data that materials used in fabricating nuts are in accordance with 3.2.

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in table II.

TABLE II. Quality conformance inspections.

| Inspection | Requirement paragraph | Test method paragraph |
|---|---|---|
| Hardness | 3.3 | 4.6.2 |
| Design | 3.5 | 4.6.1 |
| Dimensions | 3.6 | 4.6.1 |
| Discontinuities | 3.7 | 4.6.1, 4.6.3 |
| Proof load | 3.8 | 4.6.4 |
| Workmanship | 3.9 | 4.6.1 |

4.5.1 Inspection lot. An inspection lot shall consist of completed nuts covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

4.5.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.3 Sampling for visual and dimensional examination. Sampling shall be in accordance with inspection level S-3 of MIL-STD-105 with an acceptable quality level (AQL) of 2.5 percent defective for major defects. For minor defects, inspection level S-3 and an AQL of 4.0 percent defective shall apply.

4.5.4 Sampling for tests. Samples for tests shall be selected from lots which have been visually and dimensionally examined and accepted. Sampling shall be in accordance with MIL-STD-105 with inspection level and AQL as follows:

| Test | Inspection level | AQL |
|---|---|---|
| Discontinuities (magnetic particle) | S-4 | 1.5 |
| Proof load | S-4 | 1.5 |
| Hardness | S-2 | 4.0 |

4.5.5 Packaging inspection. The sampling and inspection of the preservation-packaging, packing, and container marking shall be in accordance with PPP-H-1581.

4.6 Methods of inspection.



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4.6.1 Visual and dimensional examination. Samples taken as specified in 4.5.3 shall be thoroughly examined to determine conformance with this specification and applicable specification sheet. Examination shall be conducted in accordance with table III.

TABLE III. Classification of defects.

| Category | Defect | Inspection method |
|---|---|---|
| Critical | None defined | |
| Major | | |
| 101 | Threads not as specified (see 3.5.1). | SIE 1/ |
| 102 | Bearing surface not as specified (see 3.5.2). | SIE |
| 103 | Evidence of cracks, laps, seams, inclusions (see 3.7). | Visual and SIE |
| 104 | Pilot dimensions and concentricity not as specified (see 3.5.3, 3.6). | SIE |
| 105 | Lug height not as specified (see 3.6) | SIE |
| Minor | | |
| 201 | Nut thickness not as specified (see 3.6). | SIE |
| 202 | Other dimensions not as specified (see 3.6). | SIE |
| 203 | Workmanship not as specified (see 3.9). | Visual |

1/ Standard inspection equipment.

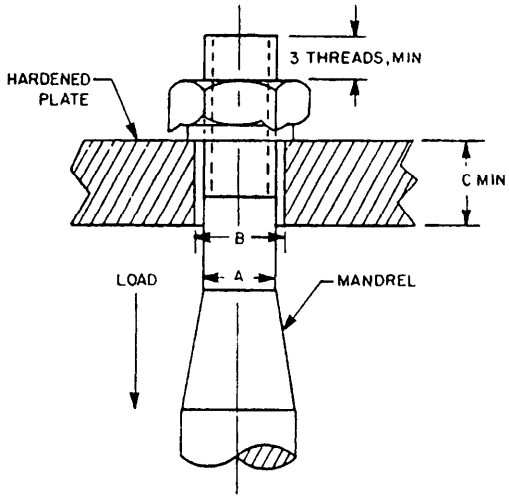
4.6.2 Hardness. When specified (see 6.2), sample nuts taken as specified in 4.5.4 shall be tested for hardness in accordance with MIL-STD-1312, test method 6.

4.6.3 Discontinuities. Nuts with thread sizes smaller than 6 mm are exempt from this test. When specified (see 6.2), sample nuts taken as specified in 4.5.4 shall be subjected to magnetic particle inspection in accordance with MIL-I-6868 for conformance to 3.7. When specified (see 6.2), samples with seams or potentially deep forging cracks shall be subjected to the cone proof load test of SAE J122A.

4.6.4 Proof load. Samples taken as specified in 4.5.4 shall be mounted on a mandrel in accordance with figure 1, and loaded as specified in 3.8, table I. Mandrel hardness shall be 45HRC, minimum, and threads shall be class 5h6g in accordance with FED-STD-H28/21, except that the maximum major diameter shall be the minimum major diameter plus 0.25 times the class 6g tolerance range. The nut shall withstand the specified load, applied axially and in tension, without rupture or stripping.



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| Dimensions | | |
|---|---|---|
| A<br>Thread size<br>(mm) | B<br><br>(mm) | C<br>min<br>(mm) |
| M3 | 3.020+0.060 | 3 |
| M4 | 4.030+0.075 | 4 |
| M5 | 5.030+0.075 | 5 |
| M6 | 6.030+0.075 | 6 |
| M8 | 8.040+0.090 | 8 |
| M10 | 10.040+0.090 | 10 |
| M12 | 12.050+0.110 | 12 |
| M14 | 14.050+0.110 | 14 |
| M16 | 16.050+0.110 | 16 |

FIGURE 1. Fixture for proof load test.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging, packing and container marking shall be in accordance with PPP-H-1581 (see 6.2).

6. NOTES

6.1 Intended use. Weld nuts are designed to be fused with the metal to which the nut is mounted. They have projections on the mating face to localize heating during welding. Weld nuts are used for blind fastening of parts. They permit use of a threaded fastener with metals too thin to be tapped, and they position readily into a drilled or punched pilot hole.



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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and the applicable specification sheet.
b. Applicable specification sheet part number.
c. Tests, if required (see 4.6.2, 4.6.3).
d. Level (degree) of protection in accordance with PPP-H-1581, ordering data (see 5.1).

6.3 Definitions.

6.3.1 Metric terms. Metric terms used in this specification are defined in ASTM E-380.

6.3.2 Quality assurance terms. Quality assurance terms used in this specification are defined in MIL-STD-109.

Custodians:
Army - AR
Air Force - 99

Review activities:
Army - AV, CR
DLA - IS
NSA - NS

User activities:
Army - ME
Navy - AS, MC

Preparing activity:
Army - AR

Agent:
DLA - IS

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